DIGITAL TWIN-ASSISTED INTRUSION DETECTION FOR IN-VEHICLE CAN NETWORKS

FINAL REVIEW • COMPLETE SYSTEM & FINAL RESULTS

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1. Introduction

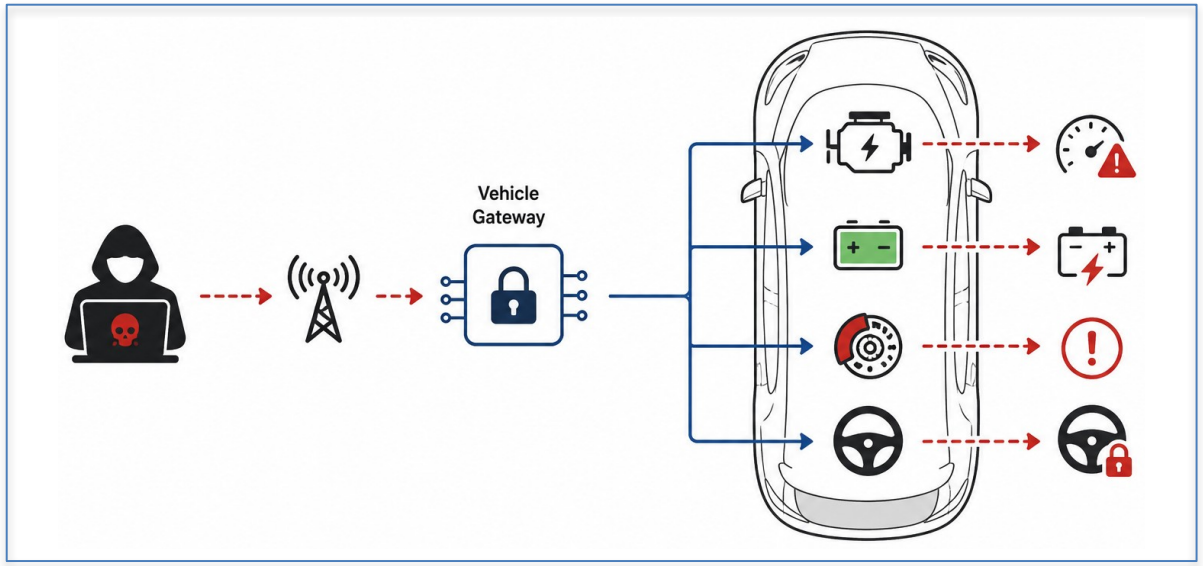
What is cyber physical attacks?

- Attacks that target the digital parts of a physical system.
- In EVs, attackers can inject or modify CAN messages.
- They can cause unsafe vehicle behaviour and system disruption

TwinShield: A Digital Twin learns normal EV CAN behaviour, finds unusual changes, combines them, and uses AI to detect and explain attacks.

Why we need to detect them?

- Protect vehicle safety by detecting malicious messages before they affect vehicle functions and prevent any disruptions.



2. Literature Review

Paper	What was done	Relevance / limitation
1. Jeong et al. (2024), IEEE Transactions on Vehicular Technology	Developed an explainable CAN intrusion detection system using meaningful CAN signals instead of only raw bits.	Provides explainability, but does not use a Digital Twin to learn expected behaviour.
2. Ding et al. (2024), Information Sciences	Developed DeepSecDrive using deep learning to detect attacks in in-vehicle networks.	Supports real-time and explainable detection, but does not use behavioural residuals.
3. Le et al. (2024), Knowledge-Based Systems	Used a Transformer and Autoencoder to detect multiple types of CAN attacks from packet and sequence information.	Strong sequence learning, but does not use a Digital Twin.
4. Kang et al. (2024), Vehicular Communications	Developed CANival by combining CAN timing and signal information for attack detection.	Uses multiple types of CAN information, but does not use Digital Twin based residual fusion.
5. Khan et al. (2024), Computers and Security	Developed DivaCAN using multiple machine learning models combined as an ensemble for CAN attack detection.	Improves detection performance, but does not model expected vehicle behaviour.

2. Literature Review

Paper	What was done	Relevance / limitation
6. Bari et al. (2023), Sensors	Compared Support Vector Machine, Decision Tree and K Nearest Neighbor models using real vehicle CAN datasets.	Uses traditional machine learning and has limited behavioral modeling
7. Rai et al. (2025), Scientific Reports	Compared LSTM, GRU, BiLSTM and other deep learning models for CAN attack detection.	Shows strong deep learning performance, but does not use Digital Twin based residual information.
8. Xu et al. (2025), PeerJ Computer Science	Developed BEPCD using feature engineering and ensemble learning for CAN intrusion detection.	Handles multiple attacks, but does not use adaptive residual fusion.
9. Wang et al. (2026), IEEE Transactions on Vehicular Technology	Used data augmentation with Generative Adversarial Networks and prediction models using CNN and LSTM for CAN attack detection.	Improves attack detection and handles limited attack data, but does not use Digital Twin residual fusion.
10. Xu et al. (2026), Internet of Things	Developed a lightweight Mamba based model for long term CAN traffic patterns and fast attack detection.	Very strong performance and low latency, but does not use Digital Twin based behavioral residuals.

2. Literature Review and Research Gap

What is already established?

- Deep learning models such as CNN, LSTM, BiLSTM and Transformer are widely used for CAN attack detection.
- Detection of attacks such as DoS, spoofing, replay and data tampering is well studied.
- SHAP and attention mechanisms are established methods for explainability and feature importance.

What is not directly covered?

- Learning normal CAN behaviour using a Digital Twin for EV intrusion detection.
- Adaptive fusion of behaviour deviations from multiple CAN or ECU streams.
- A unified framework combining Digital Twin, residual fusion, CNN-BiLSTM, and SHAP for explainable EV CAN security.

Research gap statement

Existing EV CAN intrusion detection methods mainly focus on identifying known attack patterns. **Digital Twin based behaviour deviations and adaptive multi-CAN residual fusion remain underexplored.**

3. Objective

Research focus

Develop an explainable EV CAN intrusion detection system using Digital Twin based behaviour deviations and adaptive multi-CAN residual fusion.

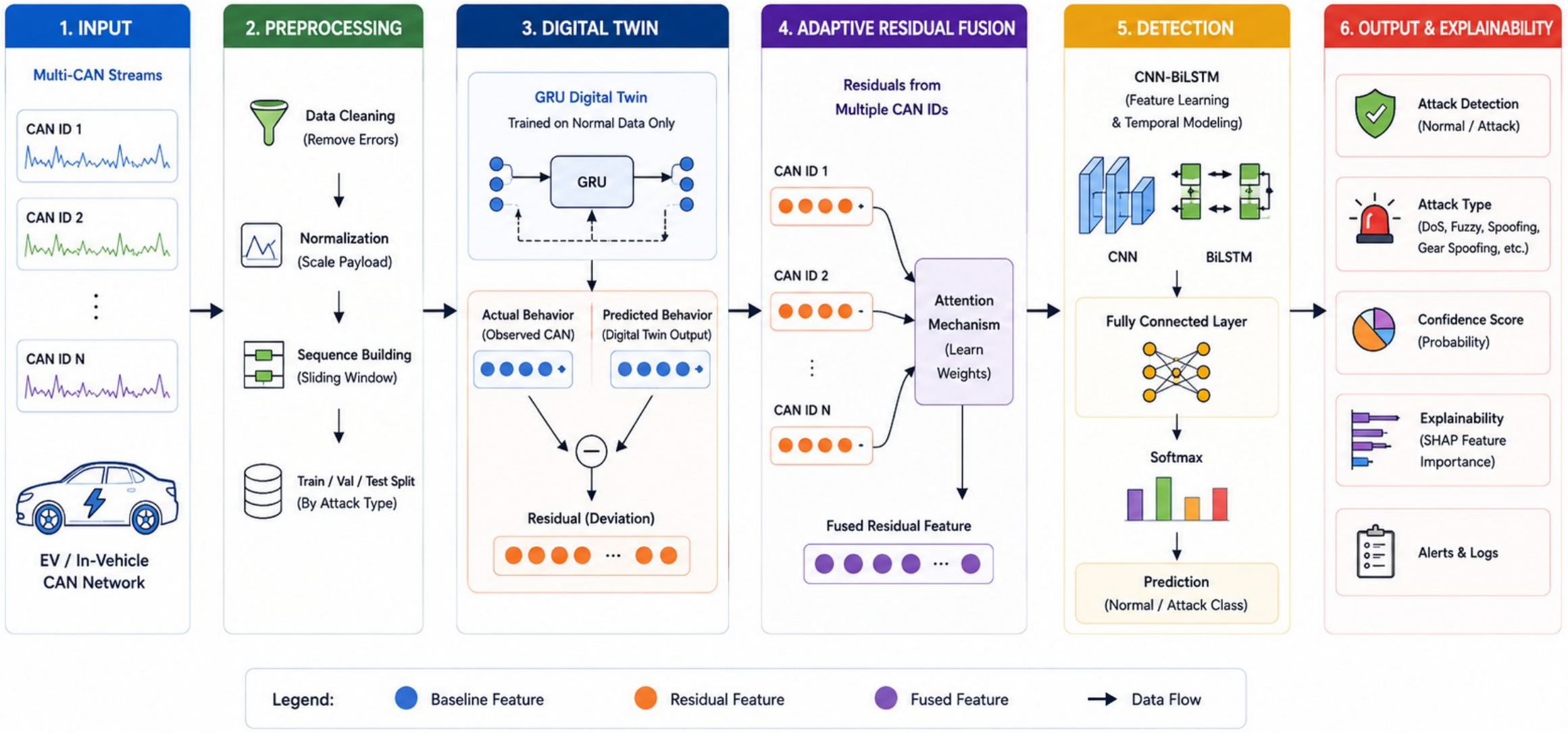
Core idea

CAN Traffic → Digital Twin → Residual Generation → Adaptive Multi-CAN Fusion → CNN-BiLSTM → Attack Detection + SHAP

Experiment

3 Baselines + 2 Proposed Variants → Compare performance using the same test data.
Evaluate the contribution of Digital Twin, residuals, and adaptive fusion.

4. Proposed Methodology



5. Implementation Status

Completed modules

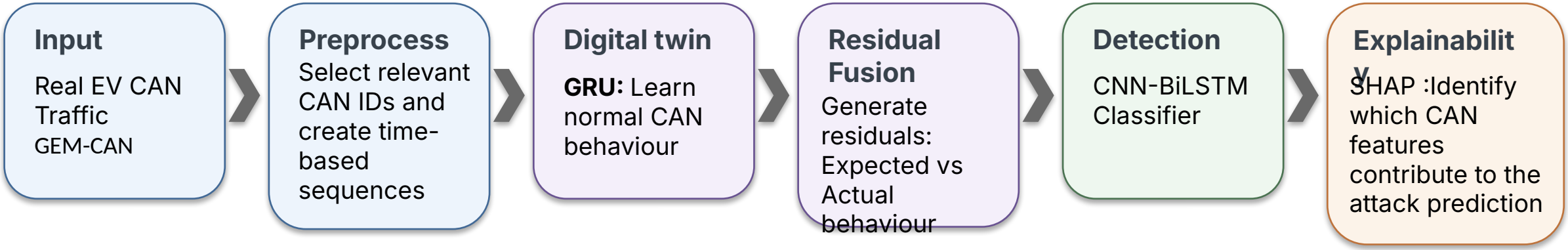
- Data pipeline: CAN parsing, windowing, splits.
- Baselines trained: CNN, BiLSTM, CNN-BiLSTM.
- GRU Digital Twin + residual generation working.
- SHAP explainability producing attributions.
- 4-page Streamlit demo (all 4 pages working).

Completed since Review 3

- Adaptive multi-ECU residual fusion trained and evaluated.
- Full ablation run, with and without the CAN ID.
- Leave-one-attack-out protocol, four-stage quarantine.
- Second dataset (SynCAN) converted and evaluated.

TwinShield: the pipeline runs end to end, every planned experiment has been run, and the report and the journal paper are complete.

6. Technical Approach and Best Practices



Best practices

- **Leakage-controlled split:** contiguous 256-frame blocks assigned whole to train/val/test, no frame's window crosses a split.
- **Class-weighted loss:** compensates for the ~152:1 Normal-vs-tampering imbalance.
- **Controlled ablation:** every model also re-run with CAN ID excluded, since ID alone can shortcut DoS.
- **Frame-level granularity:** each frame classified individually, interleaved tampering would otherwise share one label.
- **Result:** the same test split, class weights and evaluation code are shared by every model for a fair comparison.

7. Hyperparameters

Hyperparameter	Value
Sequence window	32-frame context window; 256-frame blocks for the leakage-controlled split
Input features	CAN payload bytes; CAN ID included and excluded variants
Train / Val / Test split	70 / 15 / 15 percent, block-based and stratified
GRU Digital Twin	1 GRU layer, 96 hidden units, 16-dimensional CAN-ID embedding
Digital Twin objective	Mean squared error on the next CAN payload (MAE reported as the metric)
CNN block	64 filters, kernels 5 then 3, ReLU activation
BiLSTM block	96 hidden units, bidirectional
Dropout	0.3
Classifier loss	Class-weighted cross entropy (about 152:1 imbalance)
Optimizer / Learning rate	Adam, 0.001
Batch size / Epochs	512 / 6 epochs, best validation macro-F1 checkpoint

8. Results: Baseline vs. Proposed

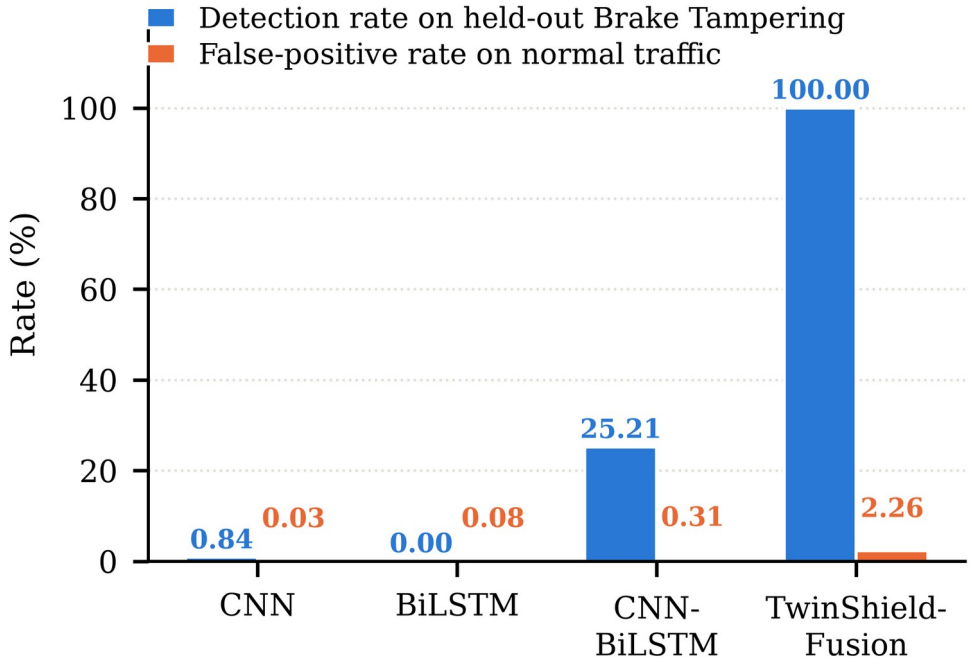
Macro-F1 on the shared held-out test split, with and without the CAN ID. The four architectures span 0.0031 macro-F1: on this benchmark accuracy no longer separates them, because a lookup table on the CAN ID alone already reaches 98.9% accuracy.

Model	CAN ID included	CAN ID excluded
CNN	0.9968	0.9914
BiLSTM	0.9948	0.9895
CNN-BiLSTM	0.9979	0.9914
GRU Digital Twin + CNN-BiLSTM	0.9989	0.9979
TwinShield-Fusion	**0.9979**	**0.9979**

9. Results: Detection of an Unseen Attack

Brake tampering is quarantined from four places: the label space, every training context window, the Digital Twin's per-ECU history and the gate's calibration set. Detection is measured on its 119 held-out test frames, false alarms on 15,063 normal test frames.

Model	Detected	False alarms	Known macro-F1
CNN	0.84%	0.03%	0.6667
BiLSTM	0.00%	0.08%	0.7607
CNN-BiLSTM	25.21%	0.31%	0.7787
TwinShield-Fusion	**100.00%**	**2.26%**	**0.7829**



10. Results: Digital Twin and Residual Evidence

Digital Twin accuracy

MAE 10.52 bytes on the 0–255 scale (0.041 normalised) predicting the next CAN payload, on held-out normal frames. Fitted on normal traffic only, so it never sees an attack pattern.

Residual separation

Brake tampering lifts the residual on 18FF00F9 from 2.64 to 25.13 bytes, a factor of 9.5, and every frame is flagged. Steering tampering lowers it from 6.54 to 3.06, and the gate misses it.

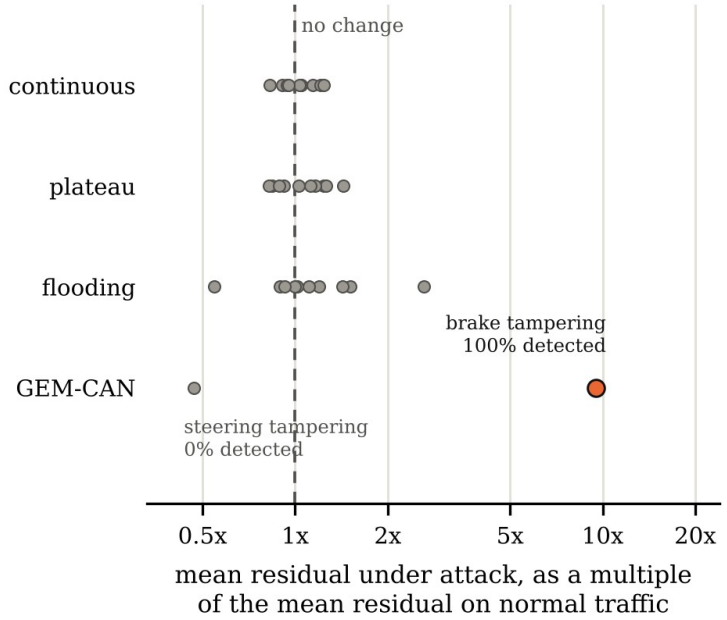
SHAP explainability

Implemented and producing per-feature attributions for every predicted class. Identifies which CAN ID and which payload byte contributed most to a detection.

11. Results: Generalisation to a Second Dataset

The whole protocol re-run on SynCAN (ETAS), 250,000 frames converted to the GEM-CAN schema. Every SynCAN attack keeps its signals inside the normal operating range, so the twin's residual barely moves and the gate does not transfer. An 8-bit control rules out the conversion: it makes the twin five times more accurate and leaves detection unchanged.

Quarantined attack	Detected	False alarms
SynCAN flooding	7.07%	2.94%
SynCAN plateau	1.56%	1.47%
SynCAN continuous	2.60%	1.45%
GEM-CAN brake tampering	**100.00%**	**2.26%**



12. Challenges and Refinements

Challenge identified

- CAN ID alone reaches near-perfect accuracy on DoS, every DoS frame uses ID 00000000, no normal frame does.
- Normal vs. tampering classes are extremely imbalanced (≈152:1).
- CAN traffic is highly repetitive, so a naive split risks train/test leakage.

How we addressed it

- Added a CAN-ID-excluded feature variant, so results aren't inflated by the ID shortcut.
- Applied class-weighted cross-entropy loss during training.
- Used block-based, stratified train/val/test splits so no frame's window crosses a split.

Key takeaway

Accuracy alone is not a meaningful headline metric on this dataset. **The residual gate earns its keep only on an attack no component was trained on, and it costs 2.26% false alarms against 0.03-0.31% for the baselines.**

13. Conclusion and Contributions

- TwinShield-Fusion is implemented end to end and evaluated on 143,444 real GEM-CAN traffic.
- Closed-set accuracy separates the four architectures by only 0.0031 macro-F1; under a leave-one-attack-out quarantine they separate by 74 percentage points, and the whole of that detection comes from the Digital Twin residual gate, not from the supervised head.
- The method has a measured boundary: it detects attacks that put implausible values on the bus and not attacks that put plausible ones, confirmed on a second dataset.
- Deliverables: the Streamlit demo, the project report and an IEEE Access-format paper. Future work: more vehicles and drive cycles, and a timing or sequence model to cover the attacks the residual cannot see.

14. Progress Overview

Progress against the approved 6-phase plan, as of the final review

Phase 1: Done	**Dataset verification**	Verify GEM-CAN data, labels, CAN IDs, attack classes and data quality.
Phase 2: Done	**Preprocessing pipeline**	Clean, normalize and prepare CAN sequences for model training.
Phase 3: Done	**Baseline implementation**	Implement CNN, BiLSTM and CNN-BiLSTM baseline models.
Phase 4: Done	**Digital Twin Development**	Train the GRU Digital Twin and generate behavioural deviation features.
Phase 5: Done	**Proposed Model**	Adaptive multi-ECU residual fusion implemented, tuned and evaluated. TwinShield-Fusion.
Phase 6: Done	**Final evaluation**	Baseline vs. proposed comparison, ablation, SHAP, leave-one-attack-out and a second dataset all complete; report and paper written.

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